

Michelle Ding

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Education

2025 - Present	The University of Texas at Austin PhD in Quantum Computing Advised by Scott Aaronson
2021 - 2025	The University of Texas at Austin BS in Computer Science, BS in Mathematics and Business Minor CS Turing Scholars Honors Program

Research

Sep 30, 2026	Ryan Anselm, MD, Dar Gilboa, Sabee Grewal. Optimal and Verifiable Quantum Advantages in Communication Complexity arXiv:2610.05585.
Apr 12, 2025	Undergraduate Honors Thesis: Peaked Boson Sampling: towards efficiently verifiable and NISQ-able quantum advantage Manuscript. Poster. <i>Best poster award.</i>
Jul 15, 2024	George Tsoukalas, Jasper Lee, John Jennings, Jimmy Xin, MD, Michael Jennings, Amitayush Thakur, Swarat Chaudhuri. PutnamBench: Evaluating Neural Theorem-Provers on the Putnam Mathematical Competition <i>NeurIPS 2024. ICML 2024. Best Paper Award.</i>

Teaching

Fall 2025	Teaching Assistant for Intro to Quantum Information Science: Honors at UT Austin.
Fall 2024	Undergraduate Grader for Introduction to Quantum Information Science at UT Austin.
Fall 2023	Teaching Assistant for Quantum Computing Freshman Research Initiative at UT Austin.
Spring 2023	Teaching Assistant for Quantum Computing Freshman Research Initiative at UT Austin.
Fall 2022	Teaching Assistant for Discrete Mathematics: Honors at UT Austin.

Scholarships, Grants & Fellowships

Aug 2026	Argonne National Laboratory Fellowship
Apr 2026	Texas Quantum Institute Fellowship
Aug 2025	McKinley Strahan Graduate Fellowship in Computer Science
Aug 2024	Leonard E. Huber Endowed Scholarship - \$3,000
Apr 2022	Freshman Research Initiative Summer Fellowship - \$2,000

Aug 2021	Ada Mae Janick Marshall Endowed Presidential Scholarship - \$3,500
2020	National Office of Mu Alpha Theta Summer Grant - \$2,000

Experience

May 2026 -	Quantum Computing PhD Research Intern at Pacific Northwest National Laboratory , at New York City, New York.
Aug 2026	Stabilizer one-way communication complexity. Advised by Samuel Stein. Researched maximal separations between one-way randomized and quantum communication complexity based on n -qubit stabilizer inputs and measurements.
Jun 2024 - Aug 2024	Research and Development Intern for Trend Micro at Austin, TX. Created structured output logging for internal weekly test automation framework. Relevant tools: Python, MySQL, Docker, Jira, Azure Data Studio.
Nov 2023 - Feb 2024	Research Assistant for the IFML Lab at Austin, TX. IFML Retrieval Augmented Generation Models. Advised by Adam Klivans. At the Institute for Foundations of Machine Learning at UT Austin. Experimented with LLMs, vector databases, and chatbot RAG algorithms to enhance the accuracy and reliability of generative AI models for identifying NeurIPS papers. Manuscript.
Jun 2022- Jul 2022	Research Assistant at the J.J. Pickle Quantum Optics Laboratory at Austin, TX. Quantum State Tomography Experiments on PBS. Advised by Noah Davis. At the J.J. Pickle Quantum Optics Laboratory, Austin. Used class 3B Laser systems to replicate state tomography and Bell's Inequality experiments. Manuscript.

Community Service

2025	Journal reviewer: <i>Quantum</i>
Mar 2023 - Aug 2024	Officer / Core Problem Writer for UT Programming Contest (UTPC) at UT Austin. Initiate and implemented written problem editorials which are now publicly available after every contest on Codeforces and Discord. Additionally: write problems for biweekly contests and annual UT Open contest held on the Codeforces platform.
Mar 2023 - Aug 2024	Vice President for the Mathematics Club at UT Austin. Obtained for all students access to free MAA weekly journals sponsored by the Math Department, revived Putnam practice sessions, and created the club's official website.

Skills and Certifications

2024	Editor for the upcoming book <i>Integrals and Sums Fiesta II</i> .
2020,2021	Two-time AIME Qualifier at John Foster Dulles High School.
2020	US Figure Skating, Silver Level.
Mar 2020	1st Place Winner of the American Protégé International Music Talent Competition.

Last update: October 2026